

Semester Objectives (short-term, demonstrable prototype)

1. **Consolidate all modules** (2D, 3D, physics, robotics arm, etc.) into a **single unified codebase**.
 2. **Make physics systems compatible** with each other (rigid, soft, cloth, constraints) under one engine loop.
 3. **Integrate an editor GUI** (ImGui-based “Studio”) for loading scenes, spawning objects, and inspecting properties.
 4. **Add parallelization hooks** (introduce a job system and parallelize broadphase/collision in physics).
 5. **Demonstrate AI agents (Phase 1)**: simple playtest agents that can spawn projectiles or navigate a scene.
 6. **Showcase two working demos inside the editor** (e.g., 2D cannon + cloth, 3D cloth + balls).
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Master’s Thesis Objectives (full research + product vision)

1. **Complete game engine consolidation**: robust modular architecture with physics, graphics, input, assets, and scene systems unified.
2. **Full editor implementation**: multi-panel workflow (viewport, hierarchy, inspector, profiler, agents).
3. **Advanced physics compatibility**: unified solver with constraints, cloth, rigid/soft bodies, robotics arm integration.
4. **Parallelization research & implementation**: design, implement, and evaluate parallel physics (broadphase, narrowphase, cloth, AI). Benchmark efficiency.
5. **Agentic AI integration**:
 - Autonomous agents for stress testing and playtesting.
 - Perception/action framework connected to physics world.
6. **Generative AI bridges**: texture/mesh generation pipelines integrated into the asset system.
7. **Symbiotic AI (editor assistance)**: natural language console, AI-assisted debugging, and world-building helpers.

8. **Evaluation:**

- Performance profiling of parallelized physics.
- Usability study of editor + AI features.
- Research contribution: how parallelization + agentic AI can accelerate simulation/game engine workflows.

9. **Final Deliverable:** a production-ready engine + editor (“Studio”) with working AI-assisted simulation and content creation tools.